Sven Weber

Walter Benjamin Postdoctoral Fellow - UC Berkeley

Oakland, CA, United States

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Professional Profile

Investigates how arthropod communities assemble, change, and persist across space and time, combining traditional surveys and collections with next-generation sequencing, environmental DNA, and community analysis. Published work addresses forest arthropod turnover and homogenization, plant–arthropod associations, and retrospective biodiversity monitoring. Current Berkeley work examines how topography, habitat, and invasion structure Hawaiian arthropod communities. DFG Walter Benjamin Fellow at UC Berkeley with 11 peer-reviewed publications and research mentoring across 18 thesis projects.

Academic Appointments

Walter Benjamin Postdoctoral Fellow

Berkeley, CA

EVOLAB - University of California, Berkeley

Jan 2025 - Dec 2026

- Long-term biodiversity change across the Hawaiian archipelago
  - Lead collections-informed, decade-scale biodiversity comparisons across Hawai‘i
  - Investigate web reduction and body-form variation in Hawaiian *Cyclosa* using morphological, ecological, and UCE genomic evidence
  - Study arboreal wolf spiders encountered with EVOLAB colleagues during 2025 fieldwork
  - Analyze plant-arthropod and tripartite interaction networks
  - Develop pilot studies on pathogenic and invasive nematodes in Hawai‘i
  - Funded through the DFG Walter Benjamin Programme

Postdoctoral Researcher

Berkeley, CA

EVOLAB; Essig Museum of Entomology - University of California, Berkeley

Jul 2023 - Dec 2024

- PROJECT: EDIAs (Early Detection of Invasive Arthropods)
  - Developed eDNA-based methods for terrestrial biodiversity monitoring
  - Led field campaigns in Hawai‘i
  - Initiated integrative systematic research on adaptive radiation in Hawaiian spiders
  - Curated taxonomic reference data linking Hawaiian arthropod identities to molecular detections

Researcher

Trier, Germany

Department of Environmental Sciences - Trier University

Jan 2023 - May 2023

- Academic-industry collaboration with Bayer AG
  - Designed eDNA workflows to quantify change in pollinator communities
  - Coordinated project planning across academic and industry partners

Education

PhD (Dr. rer. nat.), Natural Sciences

Trier, Germany

Department of Biogeography - Trier University

Apr 2020 - Jun 2023

- Degree conferred 23 June 2023; magna cum laude (1.17)
  - Dissertation: *An evaluation of the utility of metabarcoding natural eDNA samplers for retrospective biodiversity monitoring*
  - Developed field-to-bioinformatics workflows for plant, spider, and mussel-derived DNA
  - Supported by a German Federal Environmental Foundation (DBU) doctoral scholarship

MSc Biology

Düsseldorf and Aachen, Germany

Heinrich Heine University Düsseldorf; thesis at RWTH Aachen

Oct 2017 - Sep 2019

- Degree conferred 4 November 2019; overall grade 1.8
  - Thesis (grade 1.3): Twenty-year change in spider biodiversity across habitats near Aachen
  - Compared community composition across habitats and historical sampling periods

BSc Biology *Düsseldorf, Germany*

Heinrich Heine University Düsseldorf *Oct 2010 - Sep 2017*

- Research training
  - Thesis: *Fitness differences of Arabidopsis thaliana on heavy-metal soils*
  - Training in plant and population genetics, laboratory methods, Perl, and R

Arthropod Ecology, Evolution, and Enabling Expertise

Arthropod ecology and evolution	Hawaiian *Cyclosa* and Lycosidae, morphological identification, species boundaries, UCE datasets, ecological speciation, diversification, trophic ecology, and community change
Field and natural history	Arthropod monitoring, island field campaigns, environmental sampling, invasive-species surveys, contamination-aware workflows, and plant–arthropod interaction studies
Molecular biodiversity	eDNA/eRNA, metabarcoding, DNA and RNA extraction, PCR and qPCR, primer design, long-amplicon and marker-based sequencing
Collections-based science	Legacy collections and biological repositories, archive-derived DNA, retrospective biodiversity analysis, prospective time-capsule design, and voucher–sample–sequence traceability
Collection informatics	Voucher–sample–sequence traceability, repeat-sampling frameworks, structured metadata, reproducible pipelines, Git/GitHub, renv, Markdown, and Quarto
Quantitative ecology	R, tidyverse, ggplot2, multivariate community analysis, biodiversity time series, ecological networks, and spatial-data processing
Public scholarship	Invited and plenary talks; biodiversity activities for school, university, museum, elder-care, and disability-inclusive audiences

Peer-Reviewed Publications

Sven Weber is shown in bold; underlining marks his first or shared-first authorship; & joins equal-contribution first authors.

11. Banerjee, P., […], **Weber, S.**, […], Gillespie, R., Roderick, G., […]. **(2026)**. *Comparison of environmental DNA and bulk DNA metabarcoding for assessing terrestrial arthropod diversity across three habitat types on Guam*. *Molecular Ecology Resources* 26(5), e70172.

10. Pfau, M. J., **Weber, S.**, Kennedy, S., […], Gillespie, R. **(2025)**. *Invasive spiders and their microbiomes: Patterns of microbial variation in native and invasive species in Hawai'i*. *Ecology and Evolution* 15(10), e72175.

9. Banerjee, P., […], **Weber, S.**, […], Chen, C.-Y. **(2025)**. *Harnessing Environmental DNA to Explore Frugivorous Interactions: A Case Study in Papaya (Carica papaya) and Pineapple (Ananas comosus)*. *Environmental DNA* 7(5), e70196.

8. Junk, I. & Hans, J. & Perez-Lamarque, B., Stothut, M., **Weber, S.**, […], Krehenwinkel, H. **(2025)**. *Archived natural DNA samplers reveal four decades of biodiversity change across the tree of life*. *Nature Ecology & Evolution* 9, 1873-1884.

7. Melcher, A., **Weber, S.**, Birkhofer, K., Harms, D., […], Krehenwinkel, H. **(2024)**. *To pool or not to pool - pooled metabarcoding does not affect estimates of prey diversity in spider gut content analysis*. *Ecological Entomology*.

6. Stothut, M., Mahla, L., Backes, L., **Weber, S.**, […], Krehenwinkel, H. **(2024)**. *Recovering plant-associated arthropod communities by eDNA metabarcoding historical herbarium specimens*. *Current Biology*.

5. Gajski, D., Wolff, J., Melcher, A., **Weber, S.**, […], Krehenwinkel, H. **(2024)**. *Facilitating taxonomy and phylogenetics: An informative and cost-effective protocol integrating long amplicon PCRs and third-generation sequencing*. *Molecular Phylogenetics and Evolution* 192, 107988.

4. **Weber, S.** & Stothut, M., Mahla, L., […], Krehenwinkel, H. **(2024)**. *Plant-derived environmental DNA complements diversity estimates from traditional arthropod monitoring methods but outperforms them detecting plant-arthropod interactions*. *Molecular Ecology Resources* 24(2), e13900.

3. **Weber, S.**, Junk, I., Brink, L., Woerner, M., […], Krehenwinkel, H. **(2023)**. *Molecular diet analysis in mussels and other metazoan filter feeders and an assessment of their utility as natural eDNA samplers*. *Molecular Ecology Resources* 23(2), 471-485.

2. Krehenwinkel, H. & **Weber, S.**, Broekmann, R., Melcher, A., Hans, J., […], Veith, M. **(2022)**. *Environmental DNA from archived leaves reveals widespread temporal turnover and biotic homogenization in forest arthropod communities*. *eLife* 11, e78521.

1. Krehenwinkel, H., **Weber, S.**, Künzel, S., Kennedy, S. **(2022)**. *The bug in a teacup - monitoring arthropod-plant associations with environmental DNA from dried plant material*. *Biology Letters* 18(6), 20220091.

Manuscripts Under Review and Preprints

4. **Weber, S.** & Banerjee, P., […], Gillespie, R., Roderick, G. **(2026)**. *Extraction workflow determines marker-specific recovery and reproducibility in leaf-litter eDNA metabarcoding*. Under review at *Molecular Ecology Resources*.

3. Boren, A. M. & **Weber, S.**, […], Gillespie, R., Roderick, G., Roy, K. **(2026)**. *Emerging beetle-pathogen symbioses and their consequences for forest health: Lessons from Rapid 'Ōhi'a Death in Hawai'i*. Under review at *Conservation Biology*.

2. **Weber, S.**, Hutchins, L., […], Roderick, G. **(2026)**. *Topography structures arthropod communities revealed by leaf-derived environmental DNA on O'ahu, Hawai'i*. Under review at *Environmental DNA*.

1. **Weber, S.** & Melcher, A. C., Calaor, J., Kennedy, S. R., […], Krehenwinkel, H. **(2026)**. *Spiders as natural DNA samplers recover arthropod community diversity and biotic interactions across space and time*. bioRxiv preprint. Under review at *Molecular Ecology Resources*.

Invited and Plenary Talks

Weber, S. (09/2026). *Understanding plant-arthropod interaction networks across native and introduced plant species in Hawai'i using non-lethal terrestrial environmental DNA*. California Academy of Sciences, San Francisco, USA.

Weber, S. (05/2026). *Time series: Turning biodiversity snapshots into ecological memory*. Biodivreform Workshop, Trier University, Trier, Germany.

Weber, S. (05/2026). *Does ecological speciation drive diversification? An overview of Hawaiian Lycosidae spiders*. Lycosids of the World Workshop, Southwest University, Chongqing, China.

Weber, S. (04/2026). *eDNA-based monitoring to answer ecological questions*. Ecolunch, University of California, Berkeley, USA.

Weber, S. (10/2023). *Terrestrial environmental DNA - a chance for retrospective biomonitoring using biological archives*. Essig Museum of Entomology, University of California, Berkeley, USA.

Weber, S., Junk, I., Hans, J., Stothut, M., Mahla, L., Klein, R., Paulus, M., Veith, M., Krehenwinkel, H. (03/2023). *Environmental specimen banks and other biological archives for retrospective biodiversity studies*. LTER-D Annual Meeting, online.

Weber, S. (01/2023). *eDNA-based retrospective monitoring of canopy-associated arthropod communities*. Meeting of Environmental Specimen Banks, Trier, Germany.

Weber, S. (11/2022). *Der Käfer in der Teetasse*. Natural History Museum St. Gallen Autumn Meeting, St. Gallen, Switzerland.

Weber, S. (04/2022). *Retrospective monitoring of arthropod communities and more*. Senckenberg Natural History Museum Seminar, online.

Weber, S., Melcher, A. (12/2021). *Biodiversity assessments in the 21st century - and what spiders have to do with it*. **Plenary talk**, 6th Asian Society of Arachnology Meeting, online.

Weber, S., Melcher, A. (09/2021). *DNA-gestützte Analysen von Nahrung heimischer Spinnen*. **Plenary talk**, Arachnoweekend of the German Arachnological Society, online.

Weber, S. (08/2021). *Retrospective monitoring of plant-associated arthropod communities*. German Federal Agency for Nature Conservation, Convention on Biological Diversity, Vilm, Germany.

Weber, S. (05/2021). *Entwicklung eines Umwelt-DNA (eDNA) basierten Monitorings heimischer, pflanzenassoziierter Insektengemeinschaften*. German Federal Environmental Foundation Annual Meeting, online.

Conference Presentations

Weber, S. (11/2026). *From lava to food webs: Using spider diets to reconstruct network assembly across the Hawaiian chronosequence*. Oral presentation scheduled for the International Society of Arachnology Meeting, Kochi, India.

Weber, S. (11/2026). *Leaf-derived environmental DNA reveals arthropod community structure along an 'ōhi'a elevational gradient on O'ahu, Hawai'i*. Oral presentation scheduled for the International Barcode of Life Conference, Bangkok, Thailand.

Weber, S. (10/2026). *Leaf-derived eDNA reveals invasion-linked arthropod turnover across Hawaiian forest refugia*. Oral presentation scheduled for the Global eDNA Conference, Seattle, USA.

Weber, S., SERDP Pacific EDIAs Consortium (12/2025). *Mycangia matter: Ambrosia beetles and Rapid 'Ōhi'a Death*. Pacific Entomology and Biological Control Meeting, Honolulu, USA.

Weber, S., SERDP Pacific EDIAs Consortium (09/2024). *Littered with information - eDNA biodiversity studies enabled by leaf-litter samples from Hawaiian montane wet forests*. International Barcode of Life Conference, Belém, Brazil.

Weber, S. (11/2022). *Der Käfer in der Teetasse*. 34th West German Entomologists' Meeting, Düsseldorf, Germany.

Weber, S., Melcher, A., Stothut, M., Mahla, L., Hirschler, L., Kripp, A., Krehenwinkel, H. (09/2022). *eDNA - a new application for monitoring spider diversity?* 33rd European Congress of Arachnology, Greifswald, Germany.

Weber, S. (11/2021). *eDNA-basiertes Monitoring von Arthropoden*. 33rd West German Entomologists' Meeting, online.

Weber, S., Melcher, A., Künzel, S., Broekmann, R., Paulus, M., Klein, R., Krehenwinkel, H. (08/2021). *eDNA-based monitoring of canopy-associated arthropod communities*. 32nd European Congress of Arachnology, online.

Weber, S. (06/2020). *Long-term monitoring of canopy-associated spider communities by eDNA metabarcoding from leaves*. American Arachnological Society Meeting, online.

Fellowships, Grants, and Awards

2025 - 2026	DFG - German Research Foundation - Walter Benjamin Postdoctoral Fellowship	*€ 135,000*
2021 - 2022	DBU - German Federal Environmental Foundation - Doctoral Scholarship	*€ 43,500*
2018	ISA - International Society of Arachnology - Oscar and Jan Francke Student Research Fund	*€ 870*

Teaching Experience

Main Instructor
Trier, Germany

Trier University
2019 - 2022

- Programming and Metabarcoding Analysis in R (lecture and practical; mixed level; 2021-2022)
- Biomonitoring (lecture and practical; undergraduate; 2019-2022)
- Information Retrieval and Scientific Presentation (lecture and practical; mixed level; 2019-2022)

Teaching Assistant
Trier, Germany

Trier University
2019 - 2022

- Anatomy and Diversity of the Animal Kingdom (undergraduate laboratory)

Teaching Assistant
Düsseldorf, Germany

Heinrich Heine University Düsseldorf
2011 - 2016

- Zoology (laboratory; 2011-2016)
- Molecular Biology (lecture and laboratory; 2011-2015)
- Biology for Medical Students (laboratory; 2014-2016)

Research Mentoring and Academic Service

18 degree-thesis projects Hands-on research mentoring for 5 Honors, 7 Master's, and 6 Bachelor's projects
30+ students mentored Research design, laboratory and field methods, analysis, reproducible workflows, and writing

Institution	Student - Year	Current Position
Honors Thesis		
UC Berkeley	Nimrah Khan - 2026	Undergraduate, UC Berkeley
UC Berkeley	Rachel Chen - 2025	Researcher Assistant, UCSF and Google
UC Berkeley	Maia Supple - 2025	PhD student, UCLA
UC Berkeley	Alex M. Boren - 2024	PhD Student, Stanford
UC Berkeley	Alex A. Farrow - 2023	Research assistant, UCSF
Master's Thesis		
UC Berkeley	Tim Hörmann - 2025	Master's student, Konstanz
Trier University	Arndt Schmidt - 2023	PhD student, Trier University
Trier University	Lena Hirschler - 2022	Researcher, GAIAC
Trier University	Manuel Stothut - 2022	Researcher, German Env. Specimen Bank
Trier University	Lisa Mahla - 2022	PhD student, Trier University
Trier University	Alanah Kripp - 2022	PhD student, RPTU
Trier University	Lukas Brink - 2021	Career outside academia
Bachelor's Thesis		
Trier University	Tim Hörmann - 2022	Master's student, Konstanz
Trier University	Lilly Menzl - 2021	Master's student, Freiburg
Trier University	Ferdinand Meiß - 2021	Career outside academia
Trier University	Adrienne Opala - 2020	Career outside academia
Trier University	Dana Drück - 2020	Career outside academia
Trier University	Gianis Raubuch - 2020	Career outside academia

Editorial and Peer-Review Service

29 completed reviews 19 journals since 2021; record updated September 2026
Repeat invitations Molecular Ecology Resources (5), Environmental DNA (3), Ecography (2), Frontiers in Zoology (2), Journal of Insect Conservation (2), and Zoosystematics and Evolution (2)
Review coverage Environmental DNA, molecular ecology, biodiversity monitoring, conservation, entomology, zoology, and community ecology

Journal record (alphabetical): Arachnologische Mitteilungen (1); Aquatic Conservation: Marine and Freshwater Ecosystems (1); Biodiversity and Conservation (1); Ecological Genetics and Genomics (1); Ecography (2); Environmental DNA (3); Forests (1); Frontiers in Zoology (2); Journal of Arachnology (1); Journal of Insect Conservation (2); Life (1); Molecular Ecology (1); Molecular Ecology Resources (5); PeerJ (1); Plants (1); Scientific Reports (1); Thalassas: An International Journal of Marine Sciences (1); The Journal of Basic and Applied Zoology (1); and Zoosystematics and Evolution (2).

Public Scholarship, Outreach, and Leadership

Biodiversity outreach Germany and United States - Skype a Scientist; activities for kindergarten, school, university, museum, elder-care, and disability-inclusive audiences
American football coach Germany - Coached community and national-level youth teams through structured training, motivation, and adaptive instruction
Open-science training Germany and United States - Trained students in R, Git and GitHub, renv, Markdown and Quarto, metadata, and reproducible workflows
Conference service Global eDNA Conference, Seattle - volunteer (scheduled for October 2026)

References

Prof. Dr. Rosemary Gillespie *USA*
Postdoctoral Research Advisor *University of California Berkeley*

Prof. Dr. George Roderick *USA*
Postdoctoral Research Advisor *University of California Berkeley*

Prof. Dr. Henrik Krehenwinkel *GER*
Dissertation Advisor *Trier University*